

Saravanakumar U

Full Stack Software Developer [✉ saravanakumarsk8056@gmail.com](mailto:saravanakumarsk8056@gmail.com) | [☎ +91 9498361521](tel:+919498361521)

[!\[\]\(919a2cb85b99741a73c0c31a427236a8_img.jpg\) Kovilpatti, Tamil Nadu, India](#) [!\[\]\(c9cd5a1c35167a83f09a35036fe5dcbd_img.jpg\) \[LinkedIn\]\(#\)](#) | [!\[\]\(ae1936640fabdea8c18f922ca69733fe_img.jpg\) \[GitHub\]\(#\)](#) | [!\[\]\(e81307241bb070bc7c1be4e4328b2244_img.jpg\) \[Portfolio\]\(#\)](#)

Professional Summary

Full Stack Software Developer experienced in building scalable, efficient applications across frontend and backend. Skilled in Python (FastAPI), React.js, and modern databases like Trino and Iceberg. Proficient in DevOps tools (Docker, Kubernetes, Helm) and cloud deployments on OpenShift. Experienced in creating observability platforms, automating workflows with GitHub Actions, and applying AI/ML for anomaly detection. Passionate about developing high-quality, secure, and performant microservices with a strong focus on user experience.

Skills

- **Programming Languages:** JavaScript, TypeScript, Python
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS
- **Backend:** FastAPI, Node.js, Nest.js, Firebase, Express JS, REST APIs, Websocket, Redis, Kafka
- **Python:** FastAPI, Pytest, PySpark, PyTrino, Pylceberg, pyjwt, pandas
- **Databases:** SQL, MongoDB, Trino, Iceberg
- **Mobile App development:** Flutter, React Native, Expo
- **DevOps & Tools:** GitHub Actions, Docker, OpenShift, Kubernetes, Helm, MinIO,
- **Big Data & Analytics:** Apache Spark, Apache Iceberg, Apache Kafka, Apache Hive, Apache Superset
- **Observability & Monitoring:** OpenTelemetry (OTel), Prometheus, Grafana
- **AI/ML & NLP:** YOLO, Isolation Forest, Hugging Face, GPT-2 fine-tuning
- **Vector Databases & RAG:** Milvus, Retrieval-Augmented Generation (RAG) pipeline
- **Third Party Integration:** Razorpay, Google Maps
- **Others:** Data Mesh, Observability, Role-Based Access Control

Work Experience

Full Stack Software Developer

Zaga Open Source Pvt Ltd | Apr 2024 - Present | Kovilpatti, India

- Designed and implemented a **Data Mesh Architecture** utilizing **OpenTelemetry (OTel), Kafka, Hive, Iceberg, MinIO, Trino, and Redis** for scalable and efficient data processing.
- Developed a **real-time observability platform** by integrating **tracing, logging, and metrics collection** to monitor system health.

- Built a high-performance **data ingestion framework** using **FastAPI**, enabling seamless data flow between microservices.
- Engineered a **React-based frontend** to visualize infrastructure monitoring, system health, and analytics dashboards.
- Implemented **role-based access control (RBAC)** for data access management, leveraging **Trino and Iceberg** for secure and structured access policies.
- Led **Kubernetes, Docker, Helm, and OpenShift deployments**, automating application provisioning and ensuring **high availability and scalability**.
- Optimized **MinIO-based object storage** for handling large-scale telemetry and analytics data, improving storage efficiency and retrieval performance.
- Developed real-time **data analytics and processing pipelines** using Kafka and Trino, enabling fast query execution for observability insights.
- Developed and fine-tuned **AI/ML models** for anomaly detection using **Isolation Forest and Hugging Face**.

Software Development Engineer Intern

Icanio Technologies | Sep 2023 - Feb 2024 | Tirunelveli, India

- Developed user interfaces and components using React.js, ensuring responsiveness and a seamless user experience across different devices and browsers.
 - Contributed to backend development by building RESTful APIs, handling authentication, and optimizing database operations using FastAPI, MongoDB, and PostgreSQL.
 - Worked closely with designers and backend developers to integrate frontend components with backend services.
 - Actively participated in code reviews, team meetings, and adopted agile methodologies.
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Projects

Online Meat Delivery App

- Designed and developed a **scalable microservices-based architecture** using **FastAPI**, implementing core modules for User, Product, Order, Delivery, and Authentication with **JWT and role-based access control**.
- Built a **Flutter customer app** for product browsing, ordering, and real-time delivery tracking, along with a **React Native delivery partner app** and a **React.js admin dashboard** for operations management.

- Integrated **Razorpay** for secure online payments, implemented **real-time updates** using WebSocket and Redis, and deployed services with **Docker** on **OpenShift/Kubernetes**.
- Established **CI/CD pipelines** with GitHub Actions for automated deployment, used **PostgreSQL** for data persistence, and added **push notifications** to enhance user engagement and delivery visibility.

Data Lake Management & Data Product Platform using Apache Iceberg, Python, and Trino

Designed and developed a backend-driven data management platform that enables organizations to create, manage, and publish structured data products on top of a data lake. The application integrates **Apache Iceberg (REST backend)** for schema/catalog/table management and a custom **Python backend** for business-oriented data product workflows.

Key Features:

- Created logical data structures (domains, catalogs, schemas) through a user-friendly UI.
- Leveraged Apache Iceberg and MinIO for scalable table storage and versioned data management.
- Built a Python-based backend to manage data products and expose queries as APIs.
- Integrated **Trino** as a query engine, enabling users to explore data, build queries visually, and apply filters without writing code.
- Enabled publishing of saved queries as REST APIs to be consumed by other teams or applications.

Technologies Used: Python, FastAPI, Docker, Apache Iceberg, MinIO, Trino, PostgreSQL, REST APIs

Application Performance Monitoring (APM) System

- Implemented a robust monitoring system for tracing, logging, and collecting metrics to enhance real-time observability and debugging.
- Integrated OpenTelemetry (OTel) to collect telemetry data from various microservices and applications.
- Provided deep insights into application performance, helping teams optimize and troubleshoot issues efficiently.

Infrastructure Monitoring Platform

- Designed a system to monitor nodes, pods, events, and cluster health across Kubernetes environments.

- Developed real-time dashboards using Apache Superset for visualizing infrastructure health and usage patterns.
- Integrated Prometheus and Grafana for real-time alerting and performance analytics.

Edge Monitoring Solution

- Developed an edge monitoring framework to connect multiple edge devices, enabling visibility of pod, node, and cluster information in a unified dashboard.
- Ensured seamless integration with existing observability platforms to provide real-time insights into edge infrastructure.
- Implemented lightweight monitoring agents on edge devices to capture key performance metrics.

Sustainability Monitoring Dashboard

- Created a sustainability-focused dashboard for monitoring the resource efficiency of Kubernetes clusters, including node and pod sustainability tracking.
- Optimized resource allocation and sustainability by analyzing historical and real-time data on cluster resource consumption.
- Enhanced cloud cost efficiency by implementing intelligent scaling recommendations.

Anomaly Detection & Remediation Assistant (IsolationForest + Redis + RAG LLM)

- Engineered an **end-to-end anomaly detection and remediation assistant** for production observability, ingesting real-time metrics from APM agents via Redis and detecting anomalies using an **IsolationForest model**.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline that embeds contextual artifacts (logs, traces, configs) into a **FAISS** vector store and queries them to assist an **LLM** in generating precise root-cause analyses and remediation steps.
- Implemented a **feedback loop** where on-call engineers validate model suggestions, enabling **continuous learning** by updating thresholds and retraining the anomaly detector based on real-world feedback.
- Utilized a modern **Python-based stack** including scikit-learn, Redis, FAISS, sentence-transformers (OpenAI embeddings), and LLMs to deliver actionable observability insights and reduce incident response time.

Education

Bachelor of Engineering in Electronics and Communication Engineering

Government College of Engineering, Tirunelveli | 2019 - 2023 |
CGPA: 74%